Head Mounted Projection Display & Visual Attention: Visual attentional processing of head referenced static and dynamic displays while in motion and standing

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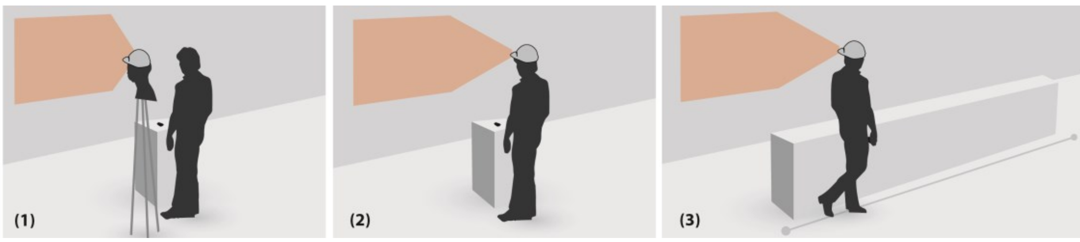

Figure 1. Experimental conditions used in the study. From left to right: (1) "control" condition representing natural viewing cases where the displayed content is referenced to world; (2) "standing" condition in which subject was immobile and wearing the HMPD prototype; (3) "walking" condition in which subject was mobile and wearing the HMPD prototype.

ABSTRACT

The Head Mounted Projection Display (HMPD) is a growing interest area in HCI. Although various aspects of HMPDs have been investigated, there is not enough information regarding the effect of HMPDs (i.e., *head referenced static* and *dynamic displays* while a user is *in motion and standing*) on visual attentional performance. For this purpose, we conducted a user study (N=18) with three experimental conditions (*control, standing, walking*) and two visual perceptual tasks (with *dynamic* and *static displays*). Significant differences between conditions were only found for the task with dynamic display; accuracy was lower in walking condition compared to the other two conditions. Our work contributes an empirical investigation of the effect of HMPDs on visual attentional performance by providing data-driven benchmarks for developing graphical user interface design guidelines for HMPDs.

Author Keywords

Head Mounted Projection Display; HMPD; Head Mounted Display; HMD; Visual Attention; Mixed Reality; Graphical User Interface; GUI; Body Mounted Projection Display; Wearable; Pico-projector; Mobile Projector, Human Factors; Ergonomics.

ACM Classification Keywords

H.5.1 Information Interfaces and Presentation: Multimedia Information Systems - *artificial, augmented, and virtual realities*; H.5.2 Information Interfaces and Presentation: User Interfaces – *ergonomics; graphical user interfaces*; H.1.2. Information Systems: User/Machine Systems – *human factors; human information processing*.

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INTRODUCTION

Body mounted projection displays are a popular field of research in HCI. The small form factor of mobile projectors makes them easily wearable. Projection displays like these are particularly promising as they allow users to access projected virtual information on physical surfaces without being connected to a desktop [10] or without an occluded vision like in the case of other Head Mounted Displays (HMDs) [2]. Researchers have tried to apply projection displays into many Mixed Reality (MR) applications by mounting them on various body parts such as wrist, shoulder, chest, hip, and head [1,2,5,7,11,13–15,18–21,24,25,27]. Among other mounting locations, the head has been an area of interest as wearing the projection display near to the eyes has key advantages. These types of technologies were discussed in previous research [9,12]. Furthermore, Bolas and Krum pointed out several advantages of these technologies while describing their near-axis retroreflective projector system, REFLCT [2]. Although HMPDs were introduced and their possible advantages over other HMDs or other projection displays were discussed, their effect on *visual attention performance*, e.g., users' performance in distinguishing, processing, and tracking visual elements shown on the screen, is still to be explored. Therefore, the aim of our research was to explore the effect of HMPDs on the visual attention performance of the users in processing static and dynamic visual elements while they are in motion or standing. More specifically, we were interested in the visual attentional processing of the *static display* properties of HMPDs where Graphical User Interface (GUI) designs contain static 2D graphical elements on the screen and the *dynamic display* properties of these devices where 2D graphical elements on the GUI are moving during different user states.

HMDs usually use two ways to present information: *head referenced* visual displays where the presented content moves with the head as in Google Glass[1] or *world referenced* visual displays where the virtual information is referenced to a point on the real environment imitating natural viewing conditions like in the case of the HoloLens[2] proposed by Microsoft (see Figure 2). Our focus, in this paper, was the head referenced GUI elements that might be used on the interfaces of HMPDs. However, the augmented reality (AR) and MR applications might also contain the combination of mentioned ways of presenting information where both world and head referenced virtual images are presented on the screen continuously [4,16]. Thus, our contribution might be beneficial for not only the applications that contain only head referenced GUI elements, but also all the applications that contain head referenced virtual graphical contents in combination with world referenced virtual images.

Despite HMDs' convenience, some researchers argue that *head referenced* visual displays might have a negative impact on visual perception, especially when the user is in motion or in a vibrating environment [6,22]. The Vestibulo-Ocular Reflex (VOR) is believed to be the primary reason

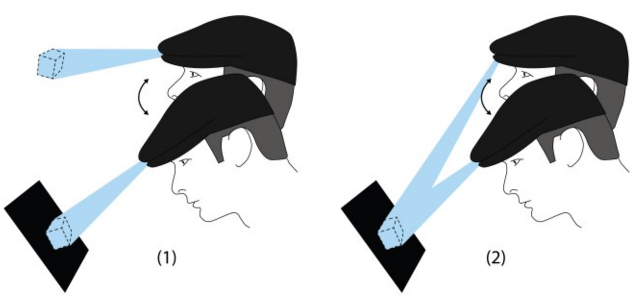

Figure 2. Two common ways to present information from an HMD: (1) Head referenced displays where the virtual object moves with the head (2) World referenced displays where the virtual information is fixed on a point in real world.

[1] www.google.com/glass/

[2] www.microsoft.com/microsoft-hololens/en-us

behind this detrimental effect. In natural settings, the VOR helps the image to be stabilized on the retina during head movements. While viewing an image displayed on a HMD, on the other hand, this reflex becomes redundant. The impact of the VOR was investigated by previous studies by looking at the effect of HMDs on visual perception during head movements caused by the mobility of the users (standing, walking or running) or environmental vibration. However, there is not sufficient information regarding the effect of HMPDs on their users' visual attention performance. The primary goal of this work is to elucidate this relation, which is directly relevant to the usability of HMPDs in daily life.

Previous research in this domain (1) primarily focused on static displays omitting the fact that dynamic displays are relevant for HMPDs in daily life or (2) they are insufficient for elucidating the effect of multiple moving visual objects on visual attention performance and (3) they focused on HMDs instead of HMPDs. Furthermore, they fail to provide data-driven benchmarks of use cases. This situation prevents us from drawing conclusions regarding the usability issues related to the cognitive performance of the users.

Our aim, therefore, is to better explore GUI design space in relation to the effects of HMPDs on attention in the context of dynamic and static displays and mobility states of humans in daily life. For this purpose, we conducted a user study (N=18), in which two visual perceptual tasks commonly-utilized in psychology literature (*Multiple Object Tracking [MOT]* for imitating dynamic contents and *Visual Search [VS]* for imitating static contents) were used for understanding the effect of display properties of HMPDs on visual attention during different user states.

There were three test conditions (see Figure 1) representing the potential states of mobility of the user in daily life while using HMPDs: (1) a *control condition* where participants perform the task while the prototype was placed on a stable surface; (2) a *standing condition* where participants performed the task while wearing an HMPD prototype; and (3) a *walking* condition in which participant performed the task while wearing the prototype.

The results of our work inform the GUI designers of the majority of AR and MR applications regarding the usability of HMPDs with dynamic and static display properties in walking and standing conditions based on their effects on visual attention performance.

BACKGROUND AND RELATED WORK

The HMPD, among other HMDs, is a rising and promising topic of research. Versions of these kinds of displays were introduced previously. Kijima and Ojika proposed a Projective Head Mounted Display (PHMD) where they used a combination of a helmet with a small LCD projector and several half mirrors[12]. In their paper, they outlined the advantages of their technology over other HMDs as being less exhaustive for the eye and being tolerable for the incorrect placement of their display on the head. Another early design of HMPDs was presented by Hue et al. [9]. Different from Kijima and Ojika, they proposed a setup using a projective lens and a beam splitter. They also explored the advantages of using retroreflective screens positioned around the real world to project virtual images on. They noted that the use of retroreflective materials as projection surfaces increase the brightness of the displays. Although both studies were valuable for realizing the capabilities of HMPDs, their displays were bulky and needed optical parts in front of the users' eyes. More recent versions of these displays typically use a pico-projector which is small in size and provide about 20-30 lumens of light output, which provides sufficient brightness when viewed using a retroreflective screen even in high-brightness ambient conditions [8]. These features of pico-projectors allow them to be used in wearable devices to create Mixed Reality (MR) experiences.

Previous work on HMPDs points out several potential benefits on visual attention. Krum et al. proposed an HMPD system called REFLCT (Retro-reflective Environments For Learner-Centered Training) for applying this technology to military training simulations [14]. They demonstrated this system to military experts and noted that HMPDs provided more natural depth cues since the virtual information was projected directly onto the physical surfaces. Instead of focusing on predefined distances as in the case of other HMDs, HMPDs indeed make it possible for their users to visually focus on proper distances. Kade et al. also explored an application of an HMPD, together with retro-reflective screens, to be used in a motion capture environment [11]. They described the distortion-free nature of the images even on non-flat surfaces due to the projector's proximity to the eyes. Sand et al. suggested a number of applications of HMPDs that included education, shopping, and museum settings [24]. They also suggested that HMPDs are more advantageous than HMDs since, with HMPDs, the field of view is not limited by the optical elements placed on the user's face.

Despite the possible visual perceptual benefits of using HMPDs emphasized by previous studies, these studies did not address the usability of HMPDs during motion and their potential interaction with critical display properties (i.e., static vs. dynamic). Importantly, these conditions capture the majority of situations in which HMPDs would be used in daily life.

Human Mobility and Visual Performance

Humans are often mobile (e.g., walking, running, sitting in a moving vehicle) in daily life. The human eye, in these cases, has a compensatory reflex called the Vestibulo-Ocular Reflex (VOR) which is activated with head movements in order to keep visual fixation and stabilize retinal image. For instance, if a person turns his/her head to the left, the VOR causes the eyes to move to the right and consequently the image on the fovea stays stable. This reflex is assumed to prevent the distortion of functional visual perception during head movements. On the other hand, among other critical factors such as low brightness and contrast, the effect of Field of View (FOV), and the absence of accommodation-vergence synergy (causing eyes to fail to focus on virtual information which is presented close to the eyes), this reflex was claimed to be unnecessary and to lead to costs in terms of visual performance during HMD use, particularly when the user is in motion with head referenced displays [6,22].

An earlier study by Sampson tested the cognitive performance of ten ground soldiers (four female) while walking with an HMD [23]. In this study a monocular occluded HMD, presenting virtual information by occluding the real environment, was used to present three reaction time tasks in four conditions (standing, walking with no obstacles, walking with obstacles on the left side, and walking with obstacles on both sides). He found that there were no significant differences on cognitive performance between the "walking without obstacles" and "standing" conditions. In a recent study, Borg et al. studied how world-fixed (natural viewing condition) and head-fixed (head referenced display condition) displays influenced the performance of reading numerical information while standing and walking [3]. They conducted their study with fourteen participants. Although they didn't report any significant performance differences between standing and walking conditions when the information was presented via world-fixed displays, the head-fixed display was found to impair visual performance in the walking condition when it was compared to the standing condition. They also found that in walking conditions the performance was lower with the head-fixed display compared to the world-fixed display. Although their study provides valuable information regarding the effect of head referenced content presentation on visual performance during motion, they specifically focused on static visual targets presented via HMDs. Despite the aforementioned benefits of HMPDs over HMDs, there is a clear empirical gap regarding the effect of HMPDs on visual attention performance. A more comprehensive understanding of these user-related issues requires experimentally addressing different visual characteristics (static vs. dynamics) as well as user states (immobile vs. mobile) within the same study.

Several lines of research investigated the effects of moving targets presented via HMDs on visual perception in cases where head movements occur. For instance, Li et al. used a moving checkerboard displayed on a head referenced HMD to examine the perception of motion during head movements in two experiments. They had six participants in the first experiment where they stabilized head movement frequency and varied the image movement frequency. The second experiment was conducted with eight participants, in which they had constant image movement frequency and tested different head movement frequencies [17]. They concluded that during horizontal movements of the head, perceptual sensitivity to motion is decreased by about 45%. In addition, they pointed out an approximately 25% decline in perceptual sensitivity during vertical head movements. Another study was conducted with six volunteers (three females). They reported decreased performance on a task that required fixating on a moving target presented by an HMD in a vibrating environment [26]. Although these studies suggested that there might be a decline in visual attentional performance with dynamic displays, the first study specifically focused on the perception of motion, instead of goal-directed task performance and they both tested participants with a single target, again leaving much to be investigated. Our study investigated how GUI designs which contain multiple moving visual objects affect visual attention performance while HMPD users are performing goal directed tasks in motion and standing.

USER STUDY

To explore the GUI design space in relation to the visual attention performance with HMPDs, we conducted a user study that primarily focused on the effects of head referenced static and dynamic displays while in motion and standing.

The experimental conditions were defined by the *three user states* (see Figure 1), which included one control condition and two mobility states of humans in daily life: *HMPD stabilized on a fixed surface* (as a control condition, representing natural viewing cases where the displayed content is referenced to world), *Standing* (participant was immobile while wearing an HMPD prototype), and *Walking* (participant was mobile while wearing an HMPD prototype). *All factors were tested as part of a within-subjects design*. The performance was evaluated separately for each task (*Visual search [VS]* task that represented *static display properties* and *multiple object tracking [MOT]* task that represented *dynamic display properties*). The *response times* and *accuracy data* were the dependent variables common to both tasks.

The study design has been approved by Koç University's Committee on Human Research.

Participants

We recruited 18 volunteers (10 females) from the Koç University community. Participants' ages ranged from 18 to 52 years (Mean = 25, SD = 7).

The sample sizes for the current study were determined primarily based on the sample sizes of the previous studies ([3,23,26] that had sample sizes of fourteen, ten and six respectively. [17] conducted their first experiment with six and their second experiment with eight participants). Furthermore, the within-subject experimental design, which was used in the current study, has higher statistical power (due to the reduction of error variance caused by individual differences) and thus requires smaller sample sizes.

Apparatus

A portable head mounted display unit was developed in the form of a cap as shown in Figure 3. The head mounted unit was composed of three different components: (1) a stripped off pico-projector (ShowWX+ from Microvision); (2) a mini computer (Raspberry Pi-1); and (3) a battery unit.

A custom 3D printed housing was used for placement of the projector and a wireless mouse was used to enable user interaction with the display. We chose mouse as the input device since it was the fastest and the most familiar way for the participants to respond and interact with the display.

The projector used for the display was a MEMS scanner based 15 lumen pico-projector with 60 Hz refresh rate, which produces white light by combining three different laser sources (RED:643 nm GREEN:530 nm and BLUE:446 nm). The use of the laser sourced projector resulted in a focused image display for various projection screen distances. The projector operates at a WVGA resolution of 848x480 pixels and a 16:9 aspect ratio. The field of view of the projection beam was 44° and 25° in horizontal and vertical directions, respectively.

To realize a standalone HMPD execute the MOT and VS test applications and record user responses in a database, a Raspberry-Pi module was used, connected to projector through an HDMI port. Raspberry-Pi is a low cost credit card sized single board computer running Linux OS. It consists of a 700 MHz processor, 1 GB of RAM and is capable of running typical graphics intended applications. Both MOT and VS applications were developed using the Python programming language with support from open-source 2D gaming libraries (i.e., pygame[3]). User responses and other test data were stored in a local database on the minicomputer and were later accessed for analysis.

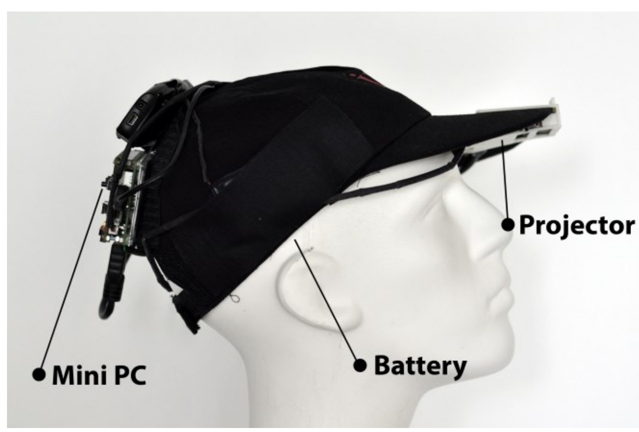

Figure 3. HMPD prototype used in the study.

[3] www.pygame.org

A typical pico-projector comes with a Lithium-ion battery with 3.7 V, 1800 mAh. In our prototype, a single power bank consisted of a pair of 3.7V, 2000 mAh Lithium-ion batteries that were used to power up pico-projector and minicomputer. The whole system worked for 1.5 to 2 hours in one charge without any interruptions.

Experiment Setting

The study took place in a university classroom, where a diffusive painted wall was used as a projection screen. The area of the wall was large enough for participants to use as display screen while standing or walking a predefined linear route in a "walking" condition. The room was dark during the sessions for the projection to be seen clearly.

A mannequin head, fixed on a tripod, was used to stabilize the HMPD in the control condition. This combination allowed for changes in HMPD height for different participants. This was important because a mismatch of height might impact the visual attention performance since the participant might look down or up due to the height difference. The mannequin head was placed 160 centimeters away from the wall.

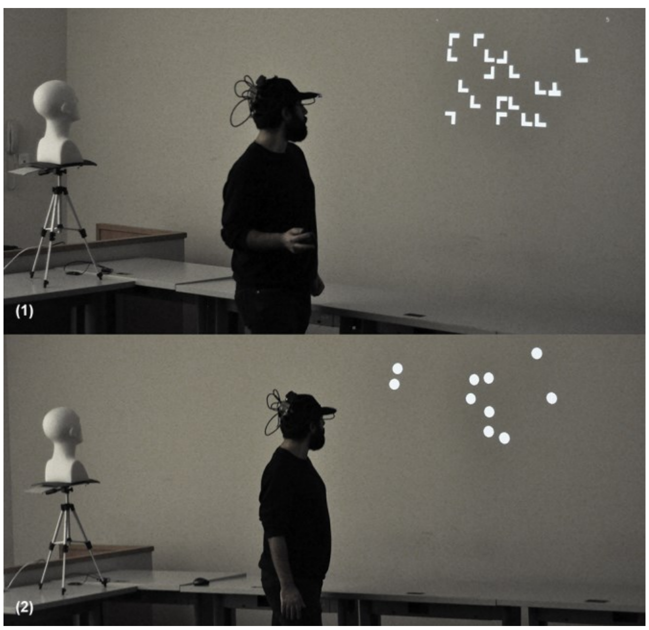

Figure 4. A user (not an actual participant) performing tasks in walking condition by following a defined path. From top to bottom: (1) the user is performing a VS task, (2) the user is performing a MOT task. The room is illuminated in these pictures to increase the visibility of the experimental setup.

A directed walking path parallel to the screen surface was structured (by using tables in the classroom) to guide the participant while walking (see Figure 4). The path was 4.5 meters without any obstacles. The path's distance from the screen surface was set to provide an approximate distance of 160 centimeters between the HMPD on the participants' head and the projection surface while participant was walking. Tables used for creating the path were also used as a surface for using the mouse. A walking condition was not implemented with the control setup due to the resultant unnatural visual angles.

Procedure

Before each experiment, participants signed an informed consent form and filled out a questionnaire on demographics. After being instructed, participants were asked to stand next to the mannequin head for the researcher to adjust the height of the mannequin to the height of the participant. Participants then began the main experiment that consisted of six blocks: control condition with MOT and VS tasks, standing condition with MOT and VS tasks, walking condition with MOT and VS tasks. Each task had a practice phase at the beginning.

In both control and standing conditions, participants were informed to stand close to the mannequin head facing the projection screen (wall). For the standing condition, participants were asked to wear the HMPD prototype and then they completed two different tasks. A mouse placed in front of them on a table was used for indicating the responses.

In the walking condition, further practice sessions were conducted to inform participants on how to respond in the tasks while walking. They were told to walk the defined path at a speed that they felt comfortable with. We chose natural walking instead of a treadmill because we were interested in natural usage scenarios of HMPDs where the participant would adjust their walking speed themselves. Additionally, by choosing natural walking, we enabled users to use the large projection screen (wall) for natural viewing experiences without limiting their viewing angle. The participants held the mouse in VS task to respond continuously while walking. In the MOT task, they indicated the response at the end of each trial by walking back to the starting point where the mouse was placed (see Figure 4). Lastly, participants were asked not to stop walking during trials.

Tasks:

Visual Search (VS) task: In a typical VS task, participants search for a target item among a number of distractor items that are randomly placed on the displayed screen. In this experiment the set size (total number of items presented in a trial) was 18 on each trial. In half of the trials, the display consisted of 17 L shaped distractor items and 1 T shaped target item (target-present). In the other trials, only distractors were presented (target-absent). The task was projected to the wall from the HMPD and all the items were 60x60 pixels in size, which corresponds to a visual angle of 3º when the projected screen was 160 cm away from HMPD (see Figure 5; right column).

In a given trial, participants responded with a right-click on the mouse to indicate that they spotted the target item, and

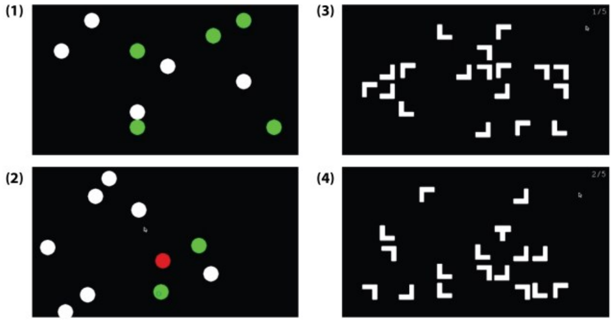

Figure 5. Illustrations of the MOT and VS tasks: (1) MOT task; targets are blinking (2) MOT task; after the targets and distractors stopped moving, subject is selecting the targets. Green indicates correct choice of the participant, red indicates incorrect choice of the participants. (3) VS task; display consists 18 L shaped distractors (4) VS task, display consists 17 L shaped distractors and 1 T shaped target.

responded with a left-click to indicate that they did not. VS is a fairly easy task that generally yields a highly accurate performance. In order to make the task more difficult to prevent a potential ceiling effect, we included a response deadline (2.5 seconds) for the trials of this task. If participants did not declare any response within 2.5 seconds, that trial ended and the next trial was started. Response times and accuracies were recorded for each trial. There were 90 trials in each condition.

Multiple Object Tracking (MOT) task: One trial of the MOT task was comprised of 10 identical discs (5 targets and 5 distractors; diameter = 36 pixels which corresponds to a visual angle of 2.4º per disc when the projected screen is 160 cm away from HMPD) scattered on the first screen (see Figure 5; left column). Five of these discs blinked several times for three seconds to mark them as targets. As soon as the blinking ceased, all discs started to move (with a speed of approximately 67 pixels per second) independently in unpredictable directions for seven seconds. Participants were instructed to identify the target discs by clicking on them with the mouse after all targets and distractors stopped moving. Every participant completed 20 trials. The inter-trial interval was eight seconds.

The whole testing procedure took one hour for each participant to perform. The order of the presentation for each condition and each task was counterbalanced between participants to prevent any possible order effect.

Data and Analysis

Repeated measures one-way ANOVA (three user states: control, standing and walking conditions) were used to analyze accuracy in both VS and MOT tasks. The accuracy data in the VS task was analyzed in two different ways; once by including trials when the response was not provided within 2.5 seconds as incorrect responses and once without including these data in the analyses. Overall, the participants failed to provide a response before the response deadline in only 3% of the trials.

Repeated measures one-way ANOVA was also performed to analyze the mean response times for VS tasks (responses later than 2.5 seconds were not included). Since the response times in the walking condition for MOT task could not be measured reliably due to the additional time participants spent to walk back to the mouse to identify targets, we did not include response times for MOT in this condition. We also omitted the response times of MOT trials where the subject failed to detect one or more targets in control and standing conditions. The response times (time taken to choose 5 circles since the discs stopped moving) collected in the standing and control conditions of this task were compared using a paired sample t-test. When the sphericity assumption was violated, degrees of freedom were corrected using the Greenhouse-Geisser estimates.

RESULTS

Response times

Figure 6 (top-right panel) shows the average response times in the VS task. The average response time was 1.52 s (SD = 0.17 s) for the control condition, 1.50 s (SD = 0.16 s) for the standing condition, and 1.54 s (SD = 0.14 s) for the walking condition. Repeated measures one-way ANOVA did not reveal a significant difference between the response times collected from three different test conditions of the VS task, $F(2, 34) = 1.65$, $p = 0.21$).

Mean response times in the MOT task did not differ between the control and standing conditions, $t(17) = 0.85$, $p = 0.41$.

Accuracy Data

For the MOT accuracy data, repeated measures one-way ANOVA revealed significant differences between the three test conditions, $F(1.41, 24.05) = 35.93$, $p<0.001$, $\eta_{p^2} = 0.68$. Participants' scores were very similar in the control ($Mean$ = 89.4%, SD = 7.4%) and standing ($Mean$ = 87.9% SD = 8.7%) conditions. By contrast, the accuracy decreased to 78.3% (SD = 11%) in the walking condition. The post-hoc comparisons between control vs. walking and standing vs. walking conditions revealed significant differences ($p<0.001$ for both comparisons). These results indicated a significant reduction in the attentional performance on the MOT task completed while walking.

The first ANOVA comparison of the accuracy data in the VS task, where the late responses were counted as incorrect responses, did not reveal a significant difference between different test conditions, $F(1.40, 23.68) = 0.26$, $p = 0.70$. The average accuracy for the VS task was 86.4% (SD = 9.6%), 87.6% (SD = 9.7%) and 86.7% (SD = 8.7%) for the control, standing, and walking conditions, respectively (see Figure 6, bottom-right panel). The second ANOVA comparison of the VS accuracy data without considering the late responses did not indicate significant differences between test conditions, either $F(1.4, 23.8) = 0.47$, $p = 0.56$.

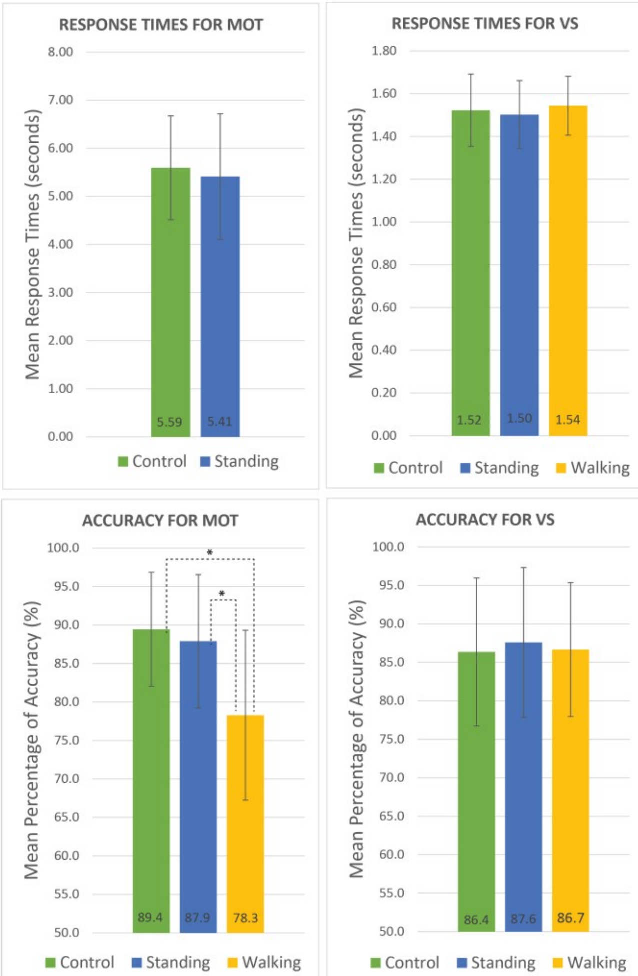

Figure 6. Mean (±SD) response times (top two panels) and accuracies (bottom two panels) for all conditions of the VS and MOT tasks. * marks significant differences between the corresponding groups.

DISCUSSION

Our results provide evidence that the critical test conditions used in our study (standing or walking) do not have an effect on the visual attention performance measured with static GUIs projected by HMPDs. This result is consistent with Sampson's findings [23] which may indicate that HMPDs do not have a detrimental effect on visual performance similar to the HMDs when GUIs are composed of static visual elements. On the other hand, our results from the static display tasks (i.e., visual search) are in contrast with the experimental findings of Borg et al.; namely decrements in reading performance while walking compared to the standing condition [3]. This discrepancy between the two sets of results might be due to the effect of HMPD usage instead of using an HMD and/or differences in task difficulties of the tasks (reading numerical information and visual search of graphical content) presented to the users. Although further research is needed to infer a more solid conclusion about the reason(s) behind this discrepancy, our results showed that static graphical contents presented via HMPD do not affect visual attentional performance. Thus, GUI designers can assign HMPD users static graphical displays in both mobile and immobile conditions without imposing detrimental effects on visual attention.

In contrast to the static displays, our results on accuracy data in the MOT task revealed significant differences between the walking and the other conditions (immobile) suggesting that presenting dynamic graphical content with HMPDs might impair visual attention while the user is in motion. In other words, participants made more errors while walking and viewing dynamic display projected by an HMPD. Our results suggest that GUI of HMPDs should avoid showing critical dynamic information or dynamic graphical elements as users might misunderstand the content and/or suffer from perceptual errors. Where possible the GUI of HMPDs should instead be composed of static graphical elements in conditions where the user is in motion.

The decreased attentional performance observed in this study should be further studied in order to clarify whether this impairment was due to HMPD usage in motion, or merely a result of walking while completing the task. On the other hand, if the walking in and by itself was indeed the source of the performance difference, one would have expected a performance difference in this condition also in the VS task, which was not the case.

Our study focused on the tasks that engaged visual attention, thus non-critical dynamic contents which are not necessary to be processed as goal-relevant visual objects might still be used in GUI of HMPDs while the user is walking. For example, visuals projected by HMPDs in museum settings could support ambient by either static or dynamic visuals while the user is wandering in the museum.

The MOT task in our study consisted of five moving targets and five distractors. Using fewer visual objects and a slower motion of the graphical elements could potentially prevent the visual attention decrements with HMPD presenting dynamic displays while in motion. Thus, further studies are needed to elucidate if slower and/or fewer moving targets also affect the visual attention performance of HMPD users during motion.

Overall, we draw a general conclusion that GUI designers working on HMPDs should consider the visual attentional effects of different designs and where possible avoid using dynamic displays when the HMPD is to be used in mobile conditions (e.g., the usage of HMPDs for navigation purposes while in motion). Arranging the GUI to be static when the user is in motion and to be dynamic when the user is immobile might increase the visual attentional

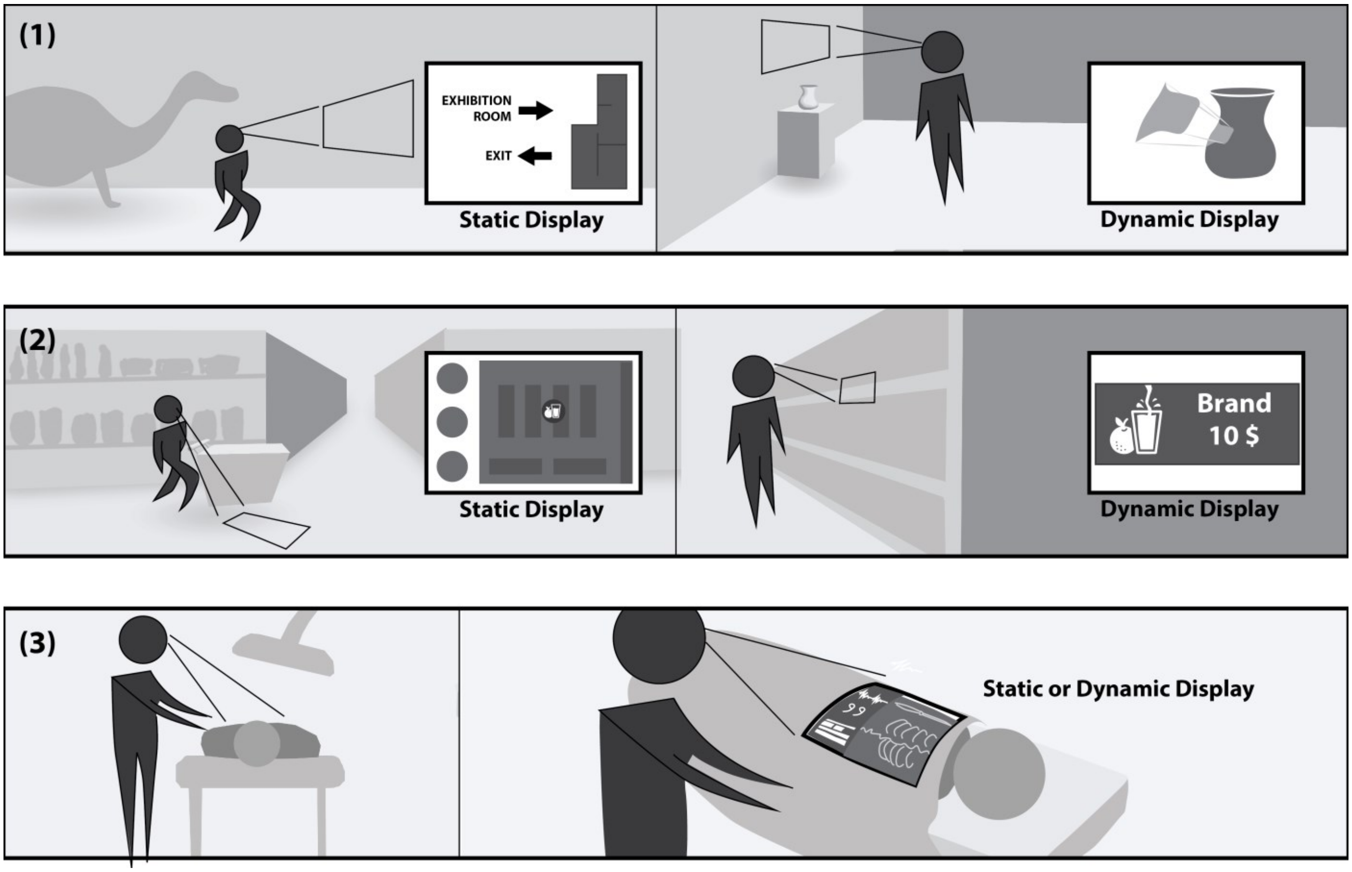

Figure 7. Examples for implementing our findings into real life scenarios: (1) Museum setting, (2) Shopping environment, (3) Surgery

performance of the user. To exemplify how our findings could be implemented in real life scenarios, we derived a few speculative explorations (see Figure 7). For example, to prevent a decrease in visual attention performance, the use of HMPDs in museum settings can include a static navigational interface while the user is wandering in the setting. A dynamic GUI design can be used when the museum visitor is in front of an artifact for presenting interactive and detailed information about the corresponding artifact/content. In another real life scenario that relates to a shopping environment, an HMPD GUI can be composed of static interfaces while the user is looking for a specific group of products (i.e., fruit juice) and a more interactive and dynamic interface could be presented when the user is looking at a specific product (i.e., particular brand or kind of fruit juice). Our results also strengthen the idea of using an HMPD in medical environments. For example, in the operation room, surgeons (who are often stable) could use an HMPD to look at a static or dynamic GUI of detailed patient information referenced to his/her head with a combination of x-ray images of the patient's body parts registered to the patient's body without any loss in visual attention performance. Although these explorations are initial ideas and they can be diversified, they demonstrate that our experimental results might provide data-driven benchmarks for constraining the GUI design space of HMPDs in different settings.

CONCLUSION

This study aimed to demonstrate the effect of different mobility conditions (including a control condition without an HMPD) on the visual attentional performance assessed with different display features presented by the HMPDs (head referenced static and dynamic displays).

In order to test the effect of HMPDs on visual attention performance, we developed an HMPD prototype and conducted a user study comparing the performance in "control", "standing" and "walking" conditions. By analyzing response time and accuracy data collected from two representative tests commonly-utilized in psychological research (VS for static display conditions, and MOT for dynamic display conditions), we investigated the effect of these critical conditions on the visual attentional performance.

The results indicate that GUIs for HMPDs can be composed of static displays while the user is in standing or walking conditions without any detrimental effects on visual attention. On the other hand, presenting dynamic displays via HMPDs might result in decrements in visual attention (i.e., accuracy), a result which should be considered by GUI designers. Finally, to exemplify how to implement our results in real life scenarios, we suggested a few initial ideas for the usage scenarios in light of our empirical findings.

Our work contributes to the HCI field by filling the empirical gap on the effect of HMPDs on visual attentional performance by providing solid data-driven benchmarks which can be used for developing GUI guidelines for HMPDs that contain head referenced GUI elements.

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